Benchmark details for "Information Budgets of Trajectory Health Indicators for Remaining Useful Life Prediction"

This document collects the benchmark material that supports Section 6 of the paper and was moved out of its appendices: the learned encoders and the predictive distribution, the data, protocols and evidence tiers, the detailed benchmark tables, and the initial benchmark protocol P1. Sections, tables and references of the paper are cited by their numbers in the paper; the bibliography is that of the paper.

S1. Estimators used in the benchmark

This appendix describes the estimators compared in Section 6. The path-space objective follows from the theory; the architectures and the calibration are design choices.

S1.1. Path-space objective

We seek a causal stochastic encoder, the *physics-guided information bottleneck* (PGIB), $q_\theta(H_{1:t} \mid \mathbf{X}_{1:t}) = \prod_{k=1}^{t} q_\theta(H_k \mid H_{k-1}, \mathbf{X}_{1:k})$, whose increments are Gaussian, $\mathcal{N}\big(H_{k-1} + u_\theta(\mathbf{X}_{1:k})\Delta t,\ s_\theta^2(\mathbf{X}_{1:k})\Delta t\big)$, with a positive velocity u_θ (enforced by a softplus output) and a variance s_θ^2. The encoder is regularised towards a physics prior $p_\phi(H_{1:t}) = \prod_k \mathcal{N}\big(H_{k-1} + \nu(H_{k-1};\phi)\Delta t,\ \sigma_B^2\Delta t\big)$, $H_0 \equiv 0$, where Δt is the time step, ν a state-dependent drift with parameters ϕ, and σ_B the diffusion coefficient.

Proposition 1 (Compression bound and path-space KL)**.** *For any prior p_ϕ, $I(\mathbf{X}_{1:t}; H_{1:t}) \le \mathbb{E}_{\mathbf{X}}\big[D_{\mathrm{KL}}(q_\theta(H_{1:t} \mid \mathbf{X}_{1:t}) \,\|\, p_\phi(H_{1:t}))\big]$, and for the Gaussian factorisation above the divergence is exactly*

$$D_{\mathrm{KL}} = \frac{1}{2}\sum_{k=1}^{t} \mathbb{E}_{q_\theta(H_{k-1}|\mathbf{X}_{1:k-1})}\left[\frac{s_\theta^2}{\sigma_B^2} - 1 - \ln\frac{s_\theta^2}{\sigma_B^2} + \frac{(u_\theta - \nu(H_{k-1};\phi))^2\,\Delta t}{\sigma_B^2}\right]. \tag{1}$$

Proof. The first claim is the standard variational upper bound of the information bottleneck [41–43]: $I(\mathbf{X}; H) = \mathbb{E}_{\mathbf{X}} D_{\mathrm{KL}}(q(H \mid \mathbf{X}) \| q(H)) \le \mathbb{E}_{\mathbf{X}} D_{\mathrm{KL}}(q(H \mid \mathbf{X}) \| p(H))$ because the difference is $D_{\mathrm{KL}}(q(H) \| p(H)) \ge 0$. The chain rule of the KL divergence [40, Sect. 2.5] over the causal factorisation gives $\sum_k \mathbb{E}_{q(H_{k-1}|\cdot)} D_{\mathrm{KL}}(q(H_k \mid H_{k-1}, \mathbf{X}_{1:k}) \| p_\phi(H_k \mid H_{k-1}))$, and the Gaussian KL with means differing by $(u_\theta - \nu)\Delta t$ and variances $s_\theta^2\Delta t$, $\sigma_B^2\Delta t$ yields the summand (the Gaussian KL of the variational autoencoder [20]). □

The training objective is $\mathcal{L}_{\mathrm{PGIB}} = \mathcal{L}_{\mathrm{FHT}} + \beta\, D_{\mathrm{KL}}$, where β weights the compression term and $\mathcal{L}_{\mathrm{FHT}}$ is the negative log-likelihood of the observed failure time under the predictive distribution below, a variational surrogate for $-I(T_t; H_{1:t})$ in the sense of [42] (Algorithm 1 in Appendix S2).

S1.2. Predictive distribution

Let the posterior of the unit's drift μ given the HI increments be $\mathcal{N}(\hat{\mu}_t, \sigma_{\mu,t}^2)$ (conjugate update). Marginalising the inverse-Gaussian distribution over μ (a Gaussian integral; random-drift Wiener models are treated in [28, 29]) gives

$$p(T_t = \tau \mid H_{1:t}) = \frac{a}{\sqrt{2\pi\tau^3(\sigma_B^2 + \sigma_{\mu,t}^2\tau)}} \exp\left(-\frac{(a - \hat{\mu}_t\tau)^2}{2\tau(\sigma_B^2 + \sigma_{\mu,t}^2\tau)}\right), \qquad a = D_{\mathrm{fail}} - \hat{D}_t. \tag{2}$$

Here σ_B^2 is the aleatoric and $\sigma_{\mu,t}^2\tau$ the epistemic contribution to the effective diffusion; as t grows $\sigma_{\mu,t}^2 \to 0$ and (2) reduces to the inverse-Gaussian distribution. It is a defective density, the missing mass being the probability that the threshold is never reached, which can happen only for a non-positive drift, and $\hat{D}_t$ is the filtered state rather than the raw HI value, which avoids the degeneracies of non-monotone paths. The distribution was checked against numerical integration of the mixture (Section 3).

S1.3. Observation-corrected and regime-aware encoders

The encoder above integrates a positive velocity, $H_k = H_{k-1} + u_\theta \Delta t + \ldots$, so nothing anchors the level to the observations (Section 4.4). The observation-corrected encoder, PGIB-KF (KF: Kalman filter), therefore adds a direct level estimate $l_k = f_l(\mathbf{h}_k)$ and a learned gain $\omega_k = \mathrm{sig}(f_\omega(\mathbf{h}_k)) \in (0,1)$ to the encoder features $\mathbf{h}_k$ (here sig is the logistic function and $\epsilon_k \sim \mathcal{N}(0,1)$ the encoder noise),

$$H_k = (1-\omega_k)\big(H_{k-1} + u_k + s_k\epsilon_k\big) + \omega_k\, l_k. \tag{3}$$

Given H_{k-1} this is again a Gaussian transition with drift $u_k^{\mathrm{eff}} = (1-\omega_k)u_k + \omega_k(l_k - H_{k-1})$ and standard deviation $s_k^{\mathrm{eff}} = (1-\omega_k)s_k$, so Proposition 1 applies unchanged: the bottleneck now prices the use of the observation beyond what the physics prior allows ($\beta \to \infty$ gives $\omega \to 0$, a pure integrator). The HI is no longer monotone by construction.

The error accumulation of an integrating encoder and its correction by the gain ω are analysed in Proposition 2.

By Proposition 1, a regime-blind estimator has an error floor, and per-regime normalisation removes the offsets but not the gains. The regime-aware variant, *PGIB-KFR*, therefore places the regime in the *observation* model, where the proposition locates it. The process model is shared, whereas the level estimate l_k and the gain ω_k use regime-specific linear heads on the encoder features, $l_k = \mathbf{h}_k^\top \mathbf{W}_{r_k}^l + b_{r_k}^l$ and $\omega_k = \mathrm{sig}(\mathbf{h}_k^\top \mathbf{W}_{r_k}^\omega + b_{r_k}^\omega)$, where r_k is the regime, which is also an input of the encoder. The regimes are the clusters of the standardised operating settings. With a single regime PGIB-KFR reduces to PGIB-KF, which we check on the single-regime subsets.

S1.4. Calibrated predictive distribution with age fusion

Because the age is always known (Theorem 1(c)), the HI-based distribution $p_{\mathrm{HI}}(\tau)$, which is (2) with diffusion $c\,v_t$ and drift variance $c\,v_t/w_t$, is pooled log-linearly [70, 71] with an age-based lifetime distribution. Here v_t is the variance of the HI increments over the trailing window of length w_t for non-learned HIs and the learned σ_B^2 for the bottleneck models, and $c > 0$ is a calibration multiplier. The age-based distribution is $p_{\mathrm{age}}(\tau \mid t) = f_L(t+\tau)/S_L(t)$, the log-normal lifetime density f_L with survival function S_L [72], fitted on training lifetimes (log-scale floor 0.15) conditioned on survival to age t. The pooled distribution is

$$p(\tau) \;\propto\; p_{\mathrm{HI}}(\tau)^{w_{\mathrm{h}}}\, p_{\mathrm{age}}(\tau \mid t)^{w_{\mathrm{a}}}, \qquad (c, w_{\mathrm{h}}, w_{\mathrm{a}}) \in \mathcal{G}, \tag{4}$$

on a finite grid $\mathcal{G}$ ($c \in [10^{-2}, 10^2]$ on a log scale, ten pairs $(w_{\mathrm{h}}, w_{\mathrm{a}})$ including $(1,0)$ and $(0,1)$). The parameters minimise the negative log-likelihood of the true RUL, a strictly proper scoring rule [73], on HI paths produced *out of sample*: each HI construction is fitted on all but one training unit and applied to the left-out unit; the test unit is never used. The family contains the plain HI-based distribution and the age-only distribution, so in-sample the pooled distribution cannot do worse than either. Weights below one temper

the corresponding distribution and widen the predictive interval, which the otherwise under-covering plain distribution requires. The pooling is a calibration procedure, not a Bayes-exact fusion, which would also require conditional independence of the HI and the age given the RUL, and division by the marginal distribution of the RUL. The same predictor is used for every HI, and a variant without fusion ($w_a = 0$) is reported for each HI, so that the effects of the HI and of the fusion can be separated.

S2. Data, protocols and evidence tiers

Data. Public run-to-failure collections were used: NASA PCoE cells B0005, B0006, B0007 and B0018 [74]; CALCE CS2 cells [75]; the FEMTO-ST/PRONOSTIA bearings [76]; the NASA IMS bearings [77, 78]; and the C-MAPSS turbofan engines [79]. For batteries, the capacity is used only to define the EOL and is never an input. The EOL is the first cycle at which the rolling-median capacity falls to 75% (NASA) or 80% (CALCE) of the initial capacity; an absolute threshold of 1.4 Ah was not used because cell B0007 never reaches it. The lifetimes are 127, 73, 162 and 100 cycles (NASA) and 542, 499, 572 and 600 cycles (CS2_35 to CS2_38); the data are truncated at the EOL, and CALCE features are averaged over blocks of 4 cycles. Six per-cycle features are extracted from the charge and discharge records: constant-current time, constant-voltage time, height and voltage of the incremental-capacity peak, mean discharge temperature (NASA) or voltage drop at discharge onset (CALCE), and the discharge voltage at a fixed time (600 s for NASA, 1200 s for CALCE).

Further cells were screened by inclusion rules fixed before running any model (scripts and outputs are supplied). A NASA cell had to be at 24°C, have at least 100 discharge cycles, be discharged at a roughly constant 1.4 to 2.1 A, start between 1.5 and 2.1 Ah and reach the 75% EOL; only the four cells above qualify. Among the CALCE cells, CS2_5, 24 and 25 are partial-discharge tests, CS2_9 is discharged at a rate of 2C (twice the nominal capacity per hour), and CS2_3, 6 and 7 are short or follow other protocols, so none was used. CS2_33 and CS2_34 have the same chemistry and run to the EOL (485 and 479 cycles) but are discharged at 0.55 A; they are used only in a separate six-cell experiment, labelled protocol-heterogeneous, and the four-cell set remains the primary one.

For bearings, each run ends at failure, so EOL is the last snapshot. Condition 1 (nominally 1800 rpm, 4000 N; Bearing1_1–1_7) and condition 2 (1650 rpm, 4200 N; Bearing2_1–2_7) are evaluated separately, each with 7 units from the learning and the full test sets; condition 3 has only 3 units and was not used. The lifetimes are 2803, 871, 2375, 1428, 2463, 2448, 2259 snapshots (condition 1) and 911, 797, 1955, 751, 2311, 701, 230 snapshots (condition 2), each snapshot covering 0.1 s of 25.6 kHz acceleration every 10 s. Eight features are computed per snapshot: logarithm of the root mean square (RMS), kurtosis, crest factor and spectral entropy of the horizontal and of the vertical channel. Features are averaged over blocks of 20 (condition 1) or 8 (condition 2) snapshots, and the RUL is reported in snapshots.

Evaluation is leave-one-unit-out: each unit in turn is held out, the remaining units are used for training (three for NASA and four-cell CALCE, five for six-cell CALCE, six for each bearing condition), and metrics are computed at every step in the window 40–90% of life and pooled over the held-out units.

Further collections. FEMTO condition 3 (1500 rpm, 5000 N; Bearing3_1–3_3, with 515, 1637 and 434 snapshots, block length 10) and the NASA IMS bearings were reserved and

used only as confirmation sets. For IMS the four bearings that failed (Test 1 bearings 3 and 4, Test 2 bearing 1, Test 3 bearing 3; 2156, 2156, 984 and 6324 snapshots of 1 s at 20 kHz every 10 min) are used with one accelerometer channel each and four features per snapshot (log RMS, kurtosis, crest factor, spectral entropy), averaged over blocks of 20 snapshots. For C-MAPSS we use the single-condition subsets FD001 and FD003 (100 training engines run to failure and 100 truncated test engines with the official RUL labels each); the 14 informative sensors are standardised with training statistics. The 100 training engines are split by identifier into 80 engines to fit the HI constructions and 20 (identifier divisible by five) to select the bottleneck configuration and to calibrate the predictor; the test engines are predicted once at their last observed cycle, and 95% bootstrap intervals are computed over engines. The multi-regime subsets FD002 (260 training and 259 test engines) and FD004 (249 and 248) have six operating regimes; for them the only change is that sensors are standardised within regime, the six regimes being the clusters of the standardised operating settings found on the fitting engines (the clusters recover the six discrete settings exactly), the same preprocessing being applied to every method; the split is by identifier as above, with 208 and 200 fitting engines.

Methods. To isolate the effect of HI construction, every HI is mapped to the same scale and passed to the same calibrated predictor of Appendix S1.4. The sensor features are standardised, and each HI is mapped by an affine map $H \mapsto (H - H_{\mathrm{s}})/(H_{\mathrm{e}} - H_{\mathrm{s}})$, where H_{s} and H_{e} are the averages, over the *training* units only, of the mean of the first five and of the last five HI values of a unit (the end of life of a training unit is known); the standardisation, the PCA direction, the combination weights of the Geo-opt HI, the encoders and the affine map are all fitted on the training units and applied unchanged to the test unit, whose own end-of-life value is never used. The PGIB scale is set by its end-of-life penalty on the training units. The predictor of Appendix S1.4 then uses: the mixed inverse-Gaussian distribution (2) with the drift taken from the slope of the smoothed HI over a trailing window of $\max(10, 0.25t)$ steps, pooled with the age-based distribution. The parameters $(c, w_{\mathrm{h}}, w_{\mathrm{a}})$ are chosen on HI paths that are out of sample for the HI construction (leave-one-training-unit-out for the small sets, the 20 calibration engines for C-MAPSS). Implementation details of the compared HIs: the Geo-opt HI is found by differential evolution; the β-VAE uses a causal gated recurrent unit (GRU) encoder with a one-dimensional latent variable and a Gaussian prior; VIB is the PGIB with a zero-drift prior, that is, without physics; the PGIB has the state-dependent drift $\nu(H; \phi) = \phi_1 \exp(\phi_2 H)$ (Appendix S1.3 for PGIB-KF). For the bottleneck models, the compression weight $\beta \in \{0.03, 0.3, 3, 30\}$ and the encoder (GRU or causal feed-forward window network) are selected by inner leave-one-training-unit-out validation of the predictive negative log-likelihood, with at most three inner folds when more than five training units are available; for PGIB-KF only $\beta \in \{0.03, 0.3, 3\}$ is searched. The held-out unit is never used. Learned models use three seeds, and the other HIs are deterministic. The sensor-free reference is the age-based distribution p_{age} of Appendix S1.4. The metrics are the RMSE and the mean absolute error (MAE) of the median prediction, the prediction interval coverage probability (PICP) and normalised average width (PINAW) of the 95% interval, and the continuous ranked probability score (CRPS) [73]. For C-MAPSS the asymmetric score is $\sum_i (e^{-d_i/13} - 1)$ for $d_i < 0$ and $\sum_i (e^{d_i/10} - 1)$ for $d_i \geq 0$, summed over engines, with d_i the predicted minus the true RUL [79].

Table S1: Algorithm 1: training of the PGIB and Bayesian FHT prognostics (Appendix S1).

Input: training sequences $\{\mathbf{X}_{1:T_n}^{(n)}, T^{(n)}\}$, threshold $D_{\text{fail}} = 1$, compression weight β, step Δt. **Output:** q_{θ^*}, (ϕ^*, σ_B^*), predictive distribution (2).
1. Causal encoder (temporal convolutional network or GRU): $u_\theta, s_\theta \leftarrow \text{softplus}(\cdot)$ from $\mathbf{X}_{1:k}$.
2. Sample the path by reparametrisation: $H_k = H_{k-1} + u_\theta \Delta t + s_\theta \sqrt{\Delta t}\, \epsilon_k$, $\epsilon_k \sim \mathcal{N}(0,1)$, $H_0 = 0$.
3. Conjugate drift update from $\Delta H_{1:t}$ to obtain $(\hat{\mu}_t, \sigma_{\mu,t}^2)$.
4. $\mathcal{L}_{\text{FHT}} = -\ln p(T_t \mid H_{1:t})$ from (2); $\mathcal{L}_{\text{KL}}$ from Proposition 1.
5. Update (θ, ϕ, σ_B) with the AdamW optimiser on $\mathcal{L}_{\text{FHT}} + \beta \mathcal{L}_{\text{KL}}$ (plus a penalty on $H_{T_{\text{EOL}}} - D_{\text{fail}}$ at the EOL).

Protocol history and evidence tiers. The study proceeded in five protocols, which determine how the numbers can be read. P1 (Appendix S4) showed that the integrating encoder under-covers and that the age-only baseline is competitive. P2 added out-of-sample calibration, age fusion and a prediction grid independent of the test unit; it was frozen, with the file hash recorded, before it was run on the confirmation sets (FEMTO condition 3 and IMS). On C-MAPSS FD001, P2 showed that the integrating encoder is considerably worse than simple HIs even with 80 training engines. The diagnosis of Proposition 2 led to P3, the observation-corrected PGIB-KF, which was frozen and run once on FD003, a subset not used before, and then, with regime-wise preprocessing, once on FD002 and FD004. (A first launch used only the first 100 training engines because the identifier list was fixed in the script; it was stopped before any result was produced, and the script and the recorded hash were corrected.) P4 added the regime-aware encoder and a regime-specific ridge baseline and was run once on each subset, the choice between PGIB-KF and PGIB-KFR and of β being made on the calibration engines only; FD002 and FD004 are therefore development sets for P4, and no independent multi-regime data set was available (the successor N-CMAPSS has only six to ten units per subset). Because the differences between learned encoders and simple HIs were of the size of the seed-to-seed variability, P5 retrained the learned encoders on the four subsets with ten fresh seeds, keeping architecture, β and calibration fixed; it supersedes the three-seed results where they disagree. The evidence tiers are: (i) *independent confirmation*: C-MAPSS FD003, FD002 and FD004 for P3 (not for P4), and FEMTO condition 3 and IMS for P2 (not for P3); (ii) *development*: NASA, CALCE (four and six cells), FEMTO conditions 1 and 2 and C-MAPSS FD001, which informed the design and whose results are therefore not unbiased estimates of out-of-sample performance.

S3. Detailed benchmark results

This appendix gives the tables behind Section 6: protocol P2 on the development and confirmation sets (Tables S2 and S3), the C-MAPSS subsets (Tables S4, S5, S6 and S7), the effect of the age fusion (Table S8), an ablation of the integrating PGIB (Table S9) and the information lower bounds (Tables S10 and S11). Protocols and evidence tiers are described in Appendix S2.

Table S2: Protocol P2, development sets. RMSE, MAE and CRPS in cycles (batteries) or 10 s snapshots (bearings); mean ± standard deviation over seeds; bold: lowest RMSE.

NASA PCoE (4 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	34.4	28.7	84	2.68	21.0
PCA-HI	24.9	19.3	95	2.46	13.8
Geo-opt HI	**24.3**	20.1	93	3.22	17.6
β-VAE-HI	31.7±2.2	25.3±1.8	98±0	4.24±0.17	20.0±0.9
VIB-HI (no physics)	33.9±1.2	28.3±1.2	90±1	5.40±0.03	27.1±0.7
PGIB-HI (ens.)	34.8	29.0	93	6.51	27.3
PGIB-KF-HI (ens.)	44.3	39.4	84	4.55	29.1

CALCE CS2 (4 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	46.5	40.0	100	1.13	28.6
PCA-HI	51.3	44.0	100	1.06	30.2
Geo-opt HI	43.3	35.7	98	0.98	26.5
β-VAE-HI	49.9±0.9	41.6±1.0	100	1.09±0.00	29.5±0.4
VIB-HI (no physics)	48.3±1.4	40.6±0.9	83±4	0.85±0.00	30.8±0.8
PGIB-HI (ens.)	**38.4**	34.2	99	0.74	23.3
PGIB-KF-HI (ens.)	50.3	36.6	77	0.83	33.0

CALCE CS2 (6 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	50.6	45.5	100	1.13	30.1
PCA-HI	56.8	50.0	100	1.09	32.8
Geo-opt HI	**46.3**	43.2	100	1.02	27.6
β-VAE-HI	73.5±4.9	60.9±3.9	93±3	1.04±0.02	42.4±3.2
VIB-HI (no physics)	54.7±1.7	48.7±1.5	92±1	0.91±0.01	32.4±1.2
PGIB-HI (ens.)	48.9	44.8	90	0.87	30.8
PGIB-KF-HI (ens.)	71.9	63.6	84	0.92	45.2

FEMTO cond. 1 (7 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	**495.8**	371.4	94	3.45	303.7
PCA-HI	786.9	621.6	72	4.72	489.2
Geo-opt HI	533.2	416.8	96	3.19	311.5
β-VAE-HI	737.1±191.7	525.4±102.0	85±6	4.84±0.04	421.4±87.9
VIB-HI (no physics)	584.1±29.4	467.0±26.4	82±5	3.48±0.13	357.9±29.0
PGIB-HI (ens.)	534.7	401.8	84	4.67	340.9

FEMTO cond. 2 (7 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	**430.6**	373.5	97	10.38	278.6
PCA-HI	558.3	450.5	67	8.52	391.1
Geo-opt HI	504.9	414.7	85	11.22	396.1
β-VAE-HI	531.2±11.3	434.6±20.1	88±2	9.78±0.02	368.4±2.0
VIB-HI (no physics)	448.2±0.4	337.7±1.3	97	11.39±0.14	318.5±2.6
PGIB-HI (ens.)	449.5	333.6	97	13.97	341.2

Table S3: Protocol P2, confirmation sets (snapshots). PGIB-KF is exploratory here.

FEMTO cond. 3 (3 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	563.6	529.7	37	10.60	460.2
PCA-HI	507.5	418.4	32	6.50	398.8
Geo-opt HI	567.1	525.1	30	12.82	453.4
β-VAE-HI	**462.0±20.8**	389.5±23.6	37	8.86±0.72	363.4±20.9
VIB-HI (no physics)	505.0±0.1	438.3±0.3	37	12.55±0.00	442.5±0.2
PGIB-HI (ens.)	510.3	450.0	37	12.56	453.5
PGIB-KF-HI (ens.)	555.0	479.3	33	9.10	438.6

NASA IMS (4 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	1692.8	1503.1	83	10.78	1106.3
PCA-HI	**745.2**	605.7	96	8.05	498.0
Geo-opt HI	942.7	702.8	99	16.31	771.6
β-VAE-HI	2769.6±282.2	1324.8±242.7	84±6	10.31±0.71	1184.6±197.8
VIB-HI (no physics)	1884.6±10.5	1434.4±12.4	91	16.41±0.01	1122.5±8.8
PGIB-HI (ens.)	1623.8	1288.1	98	16.49	1017.5
PGIB-KF-HI (ens.)	1852.3	1519.8	43	12.29	1545.6

Table S4: C-MAPSS FD003 (independent confirmation set of P3; PGIB-KFR and regime-specific ridge rows are development results). RMSE: single models, mean over seeds; seed-avg.: RMSE of the seed-averaged prediction with 95% bootstrap interval over engines.

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	47.6	42.3–52.6	39.6	114536	99	1.91	28.6
PCA-HI	33.8	28.4–38.3	25.5	10747	97	1.47	19.7
Geo-opt HI	21.9	17.8–26.0	15.9	1864	100	1.19	13.1
Ridge-HI (supervised)	26.8	22.9–30.6	21.0	6147	93	1.20	16.2
Ridge-HI, regime-specific	26.8	22.8–30.6	21.0	6147	93	1.20	16.2
β-VAE-HI	40.6±1.1	34.9–45.6	32.0±0.7	29236	99	1.69	23.8±0.3
VIB-HI (no physics prior)	37.9±0.8	31.7–42.2	29.3±0.8	18680	96	2.16	24.2±0.6
PGIB-HI (GRU/window encoder)	39.9±1.0	33.9–44.4	31.2±0.4	21945	95	1.98	24.3±0.3
PGIB-HI, 3-seed ensemble	39.5	34.1–45.0	31.1	17643	95	1.98	24.1
PGIB-KF-HI	24.2±2.5	16.4–23.1	18.6±1.5	2296	94	1.19	14.8±0.4
PGIB-KF-HI, 3-seed ensemble	19.4	16.0–22.8	14.6	790	97	1.20	12.2
PGIB-KFR-HI (regime-aware)	27.4±0.1	21.8–29.3	21.2±0.3	5417	95	1.24	15.6±0.4
PGIB-KFR-HI, 3-seed ensemble	25.4	21.6–29.1	20.3	4170	97	1.24	14.6

Table S5: C-MAPSS FD001 (development set; same layout as Table S4).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	36.6	32.4–40.8	30.5	8023	99	1.18	21.1
PCA-HI	21.8	19.1–24.4	17.9	908	96	0.83	12.6
Geo-opt HI	22.0	19.5–24.3	18.0	953	96	0.83	12.7
Ridge-HI (supervised)	26.0	22.0–29.8	19.7	1606	97	0.85	14.1
Ridge-HI, regime-specific	26.0	22.0–29.8	19.7	1606	97	0.85	14.1
β-VAE-HI	22.2±0.4	19.0–24.5	18.0±0.5	875	95	0.82	12.7±0.3
VIB-HI (no physics prior)	34.2±0.3	28.8–38.3	27.1±0.7	7386	91	0.96	19.3±0.4
PGIB-HI (GRU/window encoder)	35.6±1.0	29.8–40.9	26.9±0.6	11524	94	0.90	19.6±0.4
PGIB-HI, 3-seed ensemble	35.4	29.9–40.9	26.6	10874	94	0.90	19.4
PGIB-KF-HI	27.0±2.2	18.8–27.1	20.2±1.2	2868	82	0.56	14.9±1.1
PGIB-KF-HI, 3-seed ensemble	22.9	18.6–26.9	16.6	1309	93	0.56	12.1
PGIB-KFR-HI (regime-aware)	26.9±3.6	21.8–31.0	20.2±2.6	2937	94	0.64	14.4±2.1
PGIB-KFR-HI, 3-seed ensemble	26.5	22.0–30.9	19.5	2094	96	0.64	13.8

Table S6: C-MAPSS FD002 (multi-regime; confirmation set for P3, development set for P4; same layout as Table S4).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	38.3	35.2–41.2	31.4	53131	97	0.82	21.6
PCA-HI	28.7	26.0–31.4	21.9	10331	95	0.66	15.7
Geo-opt HI	30.5	27.0–33.9	21.8	16164	98	0.59	15.2
Ridge-HI (supervised)	29.4	26.1–32.6	21.7	11935	94	0.58	15.1
Ridge-HI, regime-specific	28.4	25.3–31.5	20.9	11149	94	0.58	14.7
β-VAE-HI	30.6±0.8	27.3–33.1	24.0±0.7	11996	92	0.65	17.0±0.4
VIB-HI (no physics prior)	36.4±0.4	33.3–39.4	29.8±0.2	26118	82	0.68	21.1±0.1
PGIB-HI (GRU/window encoder)	36.8±0.2	33.6–39.4	30.1±0.5	28553	83	0.68	21.4±0.5
PGIB-HI, 3-seed ensemble	36.6	33.7–39.4	29.9	26625	82	0.68	21.3
PGIB-KF-HI	31.3±1.5	27.8–33.1	25.4±1.5	14139	81	0.79	18.6±1.1
PGIB-KF-HI, 3-seed ensemble	30.4	27.8–32.9	24.5	10983	81	0.79	18.0
PGIB-KFR-HI (regime-aware)	30.7±2.0	27.2–32.2	24.8±2.6	14802	86	0.91	18.5±1.9
PGIB-KFR-HI, 3-seed ensemble	29.6	27.0–32.1	23.7	13089	87	0.91	17.8

Table S7: C-MAPSS FD004 (multi-regime; confirmation set for P3, development set for P4; same layout as Table S4).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	54.7	50.9–58.5	46.0	611896	95	1.31	31.8
PCA-HI	44.3	40.7–48.0	35.4	104864	92	1.03	24.9
Geo-opt HI	29.7	26.6–32.7	21.9	9468	100	0.86	16.1
Ridge-HI (supervised)	30.9	28.1–33.4	23.8	11632	94	1.14	18.5
Ridge-HI, regime-specific	31.1	28.2–33.9	24.0	10794	92	1.14	18.6
β-VAE-HI	48.1±3.1	42.7–50.2	39.0±3.0	294449	93	1.17	27.7±1.9
VIB-HI (no physics prior)	51.8±1.0	47.3–56.2	41.9±0.6	697542	90	1.13	29.5±0.4
PGIB-HI (GRU/window encoder)	49.9±2.4	44.2–53.9	39.5±1.5	1143425	90	1.42	29.3±0.7
PGIB-HI, 3-seed ensemble	49.5	44.8–54.3	38.9	1085286	91	1.42	28.9
PGIB-KF-HI	42.1±2.8	37.2–43.6	33.9±1.9	241870	78	1.08	25.6±1.3
PGIB-KF-HI, 3-seed ensemble	40.6	37.3–44.0	32.9	124580	79	1.07	24.7
PGIB-KFR-HI (regime-aware)	37.1±2.5	31.4–37.2	29.4±1.5	187056	85	1.12	22.3±0.8
PGIB-KFR-HI, 3-seed ensemble	34.8	31.8–37.8	27.8	37063	85	1.12	21.0

Table S8: Effect of the age fusion: RMSE with fusion / without fusion (after recalibration with $w_{\mathrm{a}} = 0$).

HI	NASA	CALCE-4	CALCE-6	FEMTO-1	FEMTO-2	FEMTO-3	IMS
PCA-HI	25 / 18	51 / 186	57 / 184	787 / 1470	558 / 564	507 / 503	745 / 745
Geo-opt HI	24 / 32	43 / 131	46 / 126	533 / 1083	505 / 489	567 / 437	943 / 943
β-VAE-HI	32 / 26	50 / 110	73 / 270	737 / 1421	531 / 457	462 / 396	2770 / 2770
VIB-HI (no physics)	34 / 30	48 / 59	55 / 106	584 / 681	448 / 469	505 / 507	1885 / 1885
PGIB-HI (ens.)	35 / 30	38 / 35	49 / 85	535 / 754	449 / 445	510 / 509	1624 / 1624
PGIB-KF-HI (ens.)	44 / 44	50 / 59	72 / 88	–	–	555 / 556	1852 / 1852

Table S9: Ablation of the integrating PGIB on C-MAPSS FD003 (age-fused predictor; mean ± standard deviation over three seeds). PICP, PINAW: coverage and normalised width of the 95% interval; CRPS: continuous ranked probability score.

Variant	RMSE	MAE	PICP [%]	PINAW	CRPS
Full PGIB	39.9±1.0	31.2±0.4	95±1	1.98±0.03	24.3±0.3
Without bottleneck ($\beta = 0$)	39.1±1.9	30.7±1.0	96±0	1.99±0.04	23.9±0.7
Without physics prior (VIB)	37.9±0.8	29.3±0.8	96±0	2.16±0.05	24.2±0.6
Without epistemic drift variance	45.9±1.4	37.3±1.4	92±1	1.32±0.09	26.1±0.8

Table S10: Protocol P2, lower bound on the information implied by the achieved error (nats); < 0: uninformative. $h(T)$: entropy of the pooled held-out RUL.

Dataset	$h(T)$	Age-only	PCA	Geo-opt.	β-VAE	VIB	PGIB	PGIB-KF
NASA	4.36	< 0	< 0	< 0	< 0	< 0	< 0	< 0
CALCE-4	5.67	0.41	0.31	0.48	0.34	0.37	0.60	0.33
CALCE-6	5.66	0.31	0.20	0.40	< 0	0.24	0.35	< 0
FEMTO-1	7.24	< 0	< 0	< 0	< 0	< 0	< 0	–
FEMTO-2	7.04	< 0	< 0	< 0	< 0	< 0	< 0	–
FEMTO-3	6.70	< 0	< 0	< 0	< 0	< 0	< 0	< 0
IMS	8.05	< 0	0.02	< 0	< 0	< 0	< 0	< 0

Table S11: Age-conditional information lower bound (nats) under a fitted log-normal lifetime model; $\bar{h}(T \mid t)$: mean conditional entropy. A positive value for the age-only predictor shows that the model overestimates $h(T \mid t)$.

Data set	$\bar{h}(T \mid t)$	Age-only	PCA	Geo-opt.	β-VAE	VIB	PGIB
NASA	4.67	-0.28	0.04	0.07	-0.14	-0.23	-0.29
CALCE-4	5.74	0.49	0.39	0.56	0.42	0.46	0.68
CALCE-6	5.69	0.35	0.23	0.44	-0.05	0.31	0.38
FEMTO-1	7.81	0.19	-0.28	0.11	-0.15	-0.03	0.11
FEMTO-2	7.77	0.29	0.03	0.13	0.08	0.25	0.24
FEMTO-3	7.30	-0.45	-0.35	-0.46	-0.19	-0.34	-0.35
IMS	8.65	-0.20	0.62	0.39	-0.58	-0.31	-0.16

S4. Initial benchmark protocol (P1)

The first benchmark used a simpler protocol (P1). Every HI was calibrated on in-sample training paths, the plain first-hitting-time distribution (2) was used without age fusion, and the predictive grid extended to three times the held-out unit's own life, which leaks the length of the test unit into the support of the predictive distribution (removed in P2). The pilot on NASA B0005, the results below and the observation that the age-only baseline is competitive motivated P2. The P1 results are reported for completeness and are not used for the conclusions of the main text.

Pilot run. A pilot on the NASA B0005 fold, with a GRU-only encoder and $\beta \in \{0.003, 0.03, 0.3\}$, gave a PGIB RMSE of 56.3 cycles and zero interval coverage, with the selected β at the edge of the grid. The selection grid was therefore extended and the feed-forward encoder added as a candidate. Both changes are chosen per fold by inner validation, but the decision to introduce them was made after seeing a held-out result for one fold. No baseline was modified, and no change was made after the full runs reported below; the bearing experiments were run once with the final protocol.

Results under P1. The outcome depends on the data set (Tables S12 and S13, Figure S1), and the age-only baseline is needed to read it. On NASA the RMSE of the age-only baseline is 39.8 cycles; PCA-HI (17.8), the Geo-opt HI (21.9), the β-VAE (30.8) and VIB (33.3) outperform it, whereas the PGIB (47.6) does not, and its intervals cover only 42% of the true RULs. On CALCE the RMSE of the age-only baseline is 49.3; only the PGIB (33.0) outperforms it clearly, while PCA-HI, the Geo-opt HI and the β-VAE (117 to 164) are considerably worse than the age-only baseline. With the two 0.5C-discharge cells added, the PGIB keeps the lowest RMSE (50.4) but its margin over the age-only baseline

Table S12: Protocol P1, battery benchmarks, leave-one-unit-out (mean ± standard deviation over seeds). RMSE, MAE and CRPS in cycles.

NASA PCoE (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	39.8	34.7	–	–	–
PCA-HI	0.27	0.85	0.77	17.8	14.0	97	2.52	11.1
Geo-opt HI	0.21	0.93	0.95	21.9	17.0	90	2.43	12.8
β-VAE-HI	0.30±0.01	0.71±0.00	0.95±0.01	30.8±0.8	26.0±0.5	95±1	2.06±0.08	17.6±0.5
VIB-HI (no physics prior)	1.00	1.00±0.00	0.67±0.03	33.3±2.8	27.7±2.1	88±4	3.47±0.10	20.5±2.3
PGIB-HI	1.00	1.00±0.00	0.67±0.01	47.6±2.1	42.3±1.9	42±3	2.39±0.15	34.3±1.5
CALCE CS2 (4 cells, 1C/1C) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	49.3	43.8	–	–	–
PCA-HI	0.05	0.72	0.83	130.5	110.4	92	2.74	82.3
Geo-opt HI	0.39	0.95	0.98	117.2	86.8	90	3.20	69.2
β-VAE-HI	0.02±0.01	0.72±0.01	0.80±0.01	163.7±0.7	148.0±0.9	83±1	1.87±0.02	111.1±0.7
VIB-HI (no physics prior)	1.00	1.00±0.00	0.92±0.01	53.2±0.9	47.2±1.4	71±4	1.24±0.13	38.4±0.7
PGIB-HI	1.00	1.00±0.00	0.95±0.01	33.0±4.9	29.9±3.9	55±8	0.34±0.02	22.6±3.1
CALCE CS2 (6 cells, incl. 0.5C-discharge CS2_33/34) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	54.7	50.0	–	–	–
PCA-HI	0.13	0.28	0.21	191.8	172.0	50	0.99	149.6
Geo-opt HI	0.39	0.95	0.98	106.9	82.7	95	3.11	63.7
β-VAE-HI	0.05±0.02	0.52±0.10	0.33±0.04	201.6±5.3	179.9±5.0	48±0	1.46±0.18	156.1±3.1
VIB-HI (no physics prior)	1.00	1.00±0.00	0.87±0.02	69.7±6.0	54.4±8.8	90±5	2.24±0.23	44.9±4.8
PGIB-HI	1.00	1.00±0.00	0.91±0.01	50.4±4.4	44.7±5.1	69±5	0.58±0.02	33.4±4.2

Table S13: Protocol P1, FEMTO bearing benchmarks. RMSE, MAE and CRPS in 10 s snapshots.

FEMTO bearings, condition 1 (7 units) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	602.5	515.4	–	–	–
PCA-HI	0.06	0.56	0.70	784.5	663.2	93	2.80	485.7
Geo-opt HI	0.04	0.29	0.39	770.2	623.3	91	3.34	459.1
β-VAE-HI	0.07±0.02	0.58±0.01	0.73±0.01	779.8±51.1	664.4±35.6	91±3	2.69±0.13	492.4±20.6
VIB-HI (no physics prior)	1.00	1.00±0.00	0.71±0.02	570.1±56.0	471.0±50.7	78±5	3.04±0.21	365.6±44.1
PGIB-HI	1.00	1.00±0.00	0.71±0.02	524.5±45.3	423.6±37.3	75±5	2.65±0.10	336.5±31.8
FEMTO bearings, condition 2 (7 units) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	664.4	584.6	–	–	–
PCA-HI	0.04	0.43	0.17	613.9	487.6	54	2.25	433.9
Geo-opt HI	0.06	0.37	0.46	559.2	457.1	73	2.33	356.5
β-VAE-HI	0.06±0.01	0.46±0.05	0.42±0.16	539.8±31.7	439.6±24.6	70±2	2.01±0.06	348.6±26.1
VIB-HI (no physics prior)	1.00	0.99±0.00	0.39±0.02	616.8±1.5	493.8±1.5	41±1	2.62±0.23	451.1±3.2
PGIB-HI	1.00	0.99±0.00	0.46±0.02	621.8±4.3	498.2±8.1	36±2	2.25±0.01	464.3±7.9

Table S14: Protocol P1, lower bound on the information implied by the achieved error (Theorem 1(b)), in nats; < 0: uninformative. $h(T)$: entropy of the pooled held-out RUL.

Dataset	$h(T)$	PCA	Geo-opt.	β-VAE	VIB	PGIB
NASA	4.36	0.06	< 0	< 0	< 0	< 0
CALCE (4)	5.67	< 0	< 0	< 0	0.27	0.76
CALCE (6)	5.66	< 0	< 0	< 0	< 0	0.32
FEMTO c. 1	7.24	< 0	< 0	< 0	< 0	< 0
FEMTO c. 2	7.04	< 0	< 0	< 0	< 0	< 0

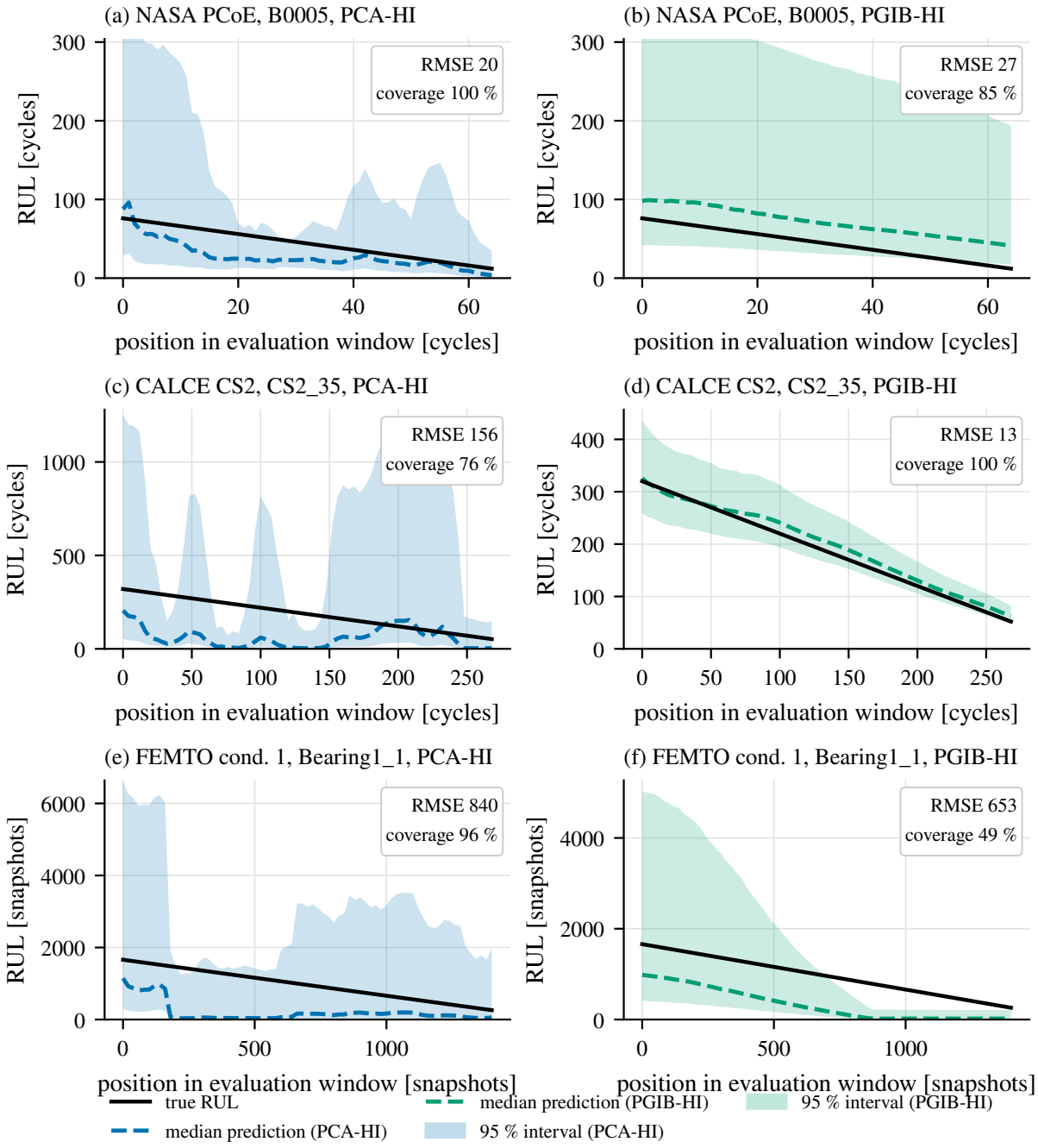


Figure S1: Example RUL predictions with 95% intervals on one held-out unit per data set (NASA B0005, CALCE CS2_35, FEMTO Bearing1_1), for PCA-HI (left) and PGIB (right); each panel gives the RMSE and the coverage on that unit.

(54.7) is only 8%. On bearings the RMSE of the age-only baseline (602 and 664 snapshots for conditions 1 and 2) is comparable to the standard deviation of the pooled RUL, and most HIs do no better: the PGIB has the lowest RMSE in condition 1 (524.5, 13% below the age-only baseline) but gives almost no benefit in condition 2 (621.8 against 664). Two long-lived units dominate the pooled error in condition 2.

The ablations (in the supplied result files) support the bottleneck on all five sets (for example RMSE 54.4 $\rightarrow$ 47.6 on NASA and 42.3 $\rightarrow$ 33.0 on CALCE) and the epistemic drift variance for coverage (for example 14 $\rightarrow$ 42% on NASA). The physics prior helps on

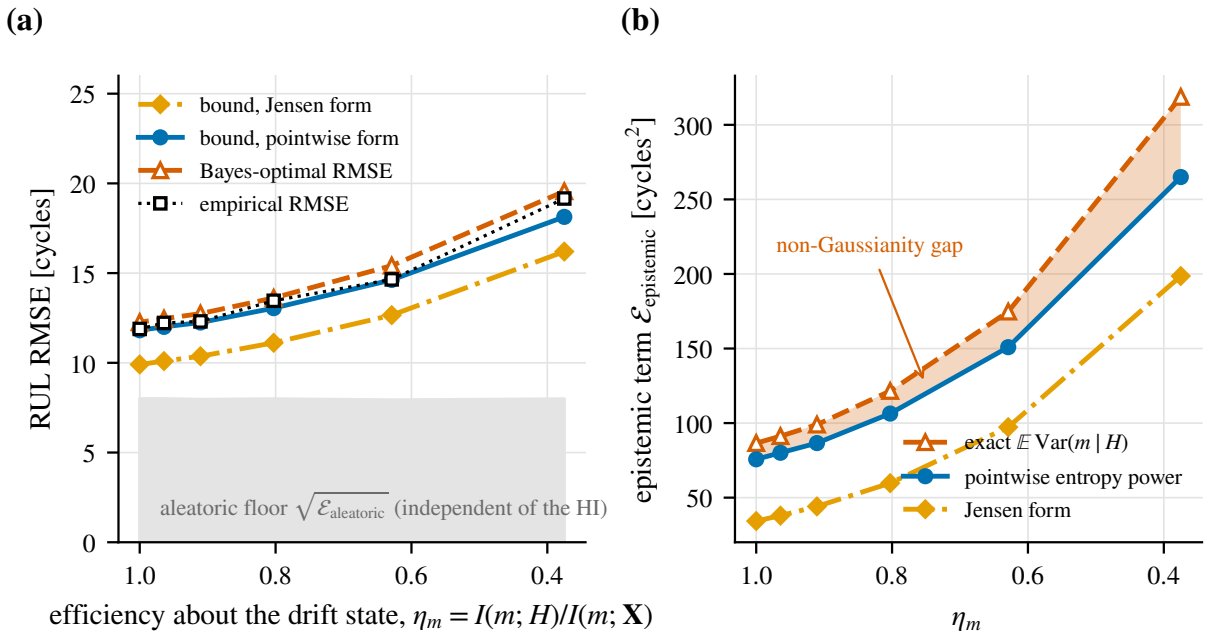


Figure S2: Physics-constrained bound as the HI loses information (decreasing η_m). (a) RMSE of the bounds and estimators. (b) Epistemic term; the shaded area is the non-Gaussianity gap.

CALCE and FEMTO condition 1 but not on NASA or FEMTO condition 2. With only three to six training units per fold, the validation signal that selects the encoder is itself noisy; on NASA it chose the strongest compression ($\beta = 30$), at the edge of the grid, in all eight selections.

Geometric scores versus error. Across the five HIs of each data set, the rank correlation between the geometric score Mon + Tre + Pro and the RMSE is +0.90 on NASA (a higher geometric score goes with a larger error), −1.00 on CALCE, −0.90 on six-cell CALCE, −0.60 on FEMTO condition 1 and +0.60 on condition 2. The geometric criteria therefore did not rank the HIs consistently by prognostic error. This is an association over five methods per data set and not evidence for the mechanism of Theorem 3: the VIB and PGIB HIs have Mon ≈ Tre ≈ 1 because their velocity is constrained to be positive, not because of a squeeze.

S5. Figures and tables moved out of the paper

These figures and tables duplicate numbers that the paper gives in its text and in other tables.

References

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Table S15: Template shrinkage of the ridge HI: monotonicity and information lower bound (nats).

Subset		$\gamma = 0.0$	$\gamma = 0.5$	$\gamma = 0.9$	$\gamma = 0.97$	$\gamma = 0.99$	$\gamma = 0.995$	Mon_∞ (predicted)
FD001	Mon	0.24	0.34	0.54	0.56	0.55	0.55	0.55
	lower bound	0.31	0.28	0.21	0.09	-0.06	-0.08	
FD002	Mon	0.19	0.28	0.54	0.58	0.58	0.58	0.61
	lower bound	0.21	0.20	0.18	0.11	0.00	-0.06	
FD003	Mon	0.16	0.22	0.40	0.41	0.42	0.42	0.41
	lower bound	0.10	0.10	0.10	0.07	0.02	-0.07	
FD004	Mon	0.13	0.18	0.36	0.41	0.41	0.41	0.41
	lower bound	0.17	0.18	0.16	0.13	0.07	0.00	
HUST	Mon	0.32	0.53	0.80	0.81	0.81	0.81	0.82
	lower bound	0.30	0.28	0.22	0.12	0.04	-0.00	
MATR	Mon	0.39	0.46	0.50	0.48	0.48	0.48	0.49
	lower bound	0.32	0.28	0.17	-0.02	-0.14	-0.20	
Tongji	Mon	0.65	0.70	0.73	0.73	0.73	0.73	0.70
	lower bound	0.62	0.54	0.33	0.16	0.09	-0.00	

Table S16: Direction of a scalar linear HI: range of stage-optimal angles, best static angle, and worst-stage RMSE ratios to the direction chosen separately at each stage.

Subset	stages	optimal angle [deg]	static angle [deg]	static / free	level only / free	velocity only / free
FD001	life fraction	43–70	70	1.019	1.18	1.07
FD001	age	33–69	60	1.021	1.20	1.18
FD002	life fraction	18–66	55	1.002	1.13	1.05
FD002	age	21–56	45	1.037	1.19	1.29
FD003	life fraction	11–33	20	1.003	1.05	1.16
FD003	age	6–35	25	1.025	1.08	1.38
FD004	life fraction	1–20	5	1.007	1.02	1.10
FD004	age	9–37	15	1.018	1.05	1.33
HUST	life fraction	3–80	70	1.090	1.13	1.10
HUST	age	0–57	20	1.184	1.08	1.54
MATR	life fraction	42–88	65	1.000	1.06	1.01
MATR	age	0–10	0	1.012	1.01	1.42
Tongji	life fraction	57–89	90	1.031	1.35	1.03
Tongji	age	13–55	40	1.068	1.12	1.18

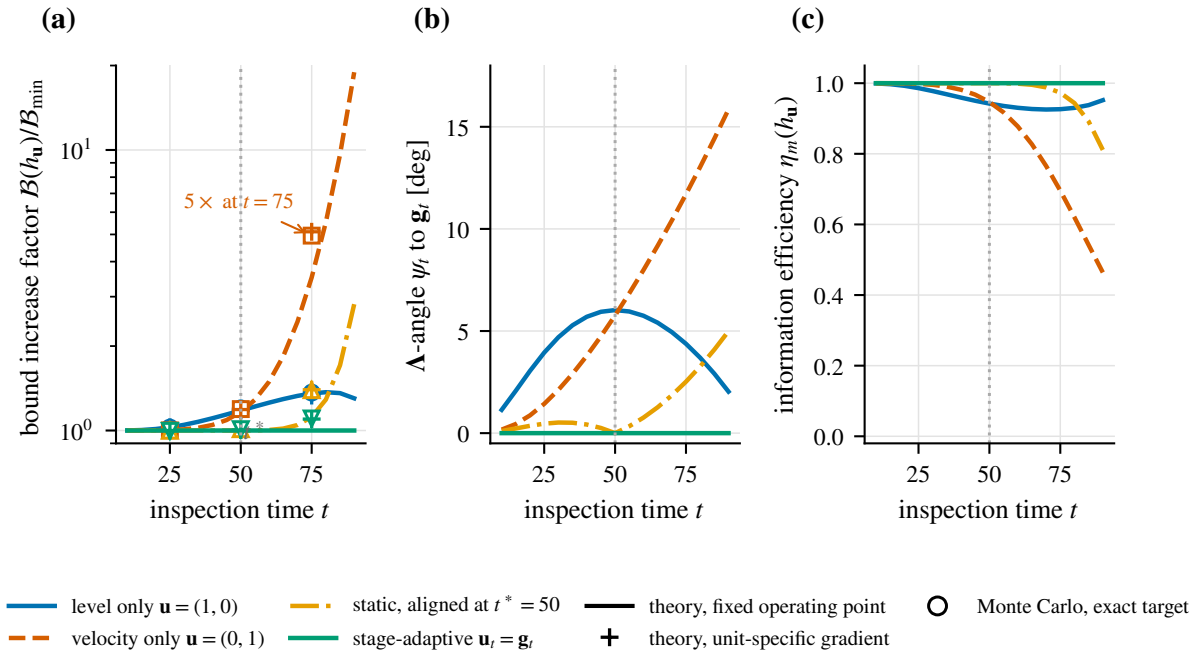


Figure S3: Angular loss relation for scalar linear HIs. (a) Factor by which the bound increases. (b) Angle to the optimal direction $\mathbf{g}_t$ at the fixed operating point. (c) Information efficiency η_m. The static HI is aligned at $t^* = 50$; the stage-adaptive HI uses the stage-wise gradient at the prior mean.

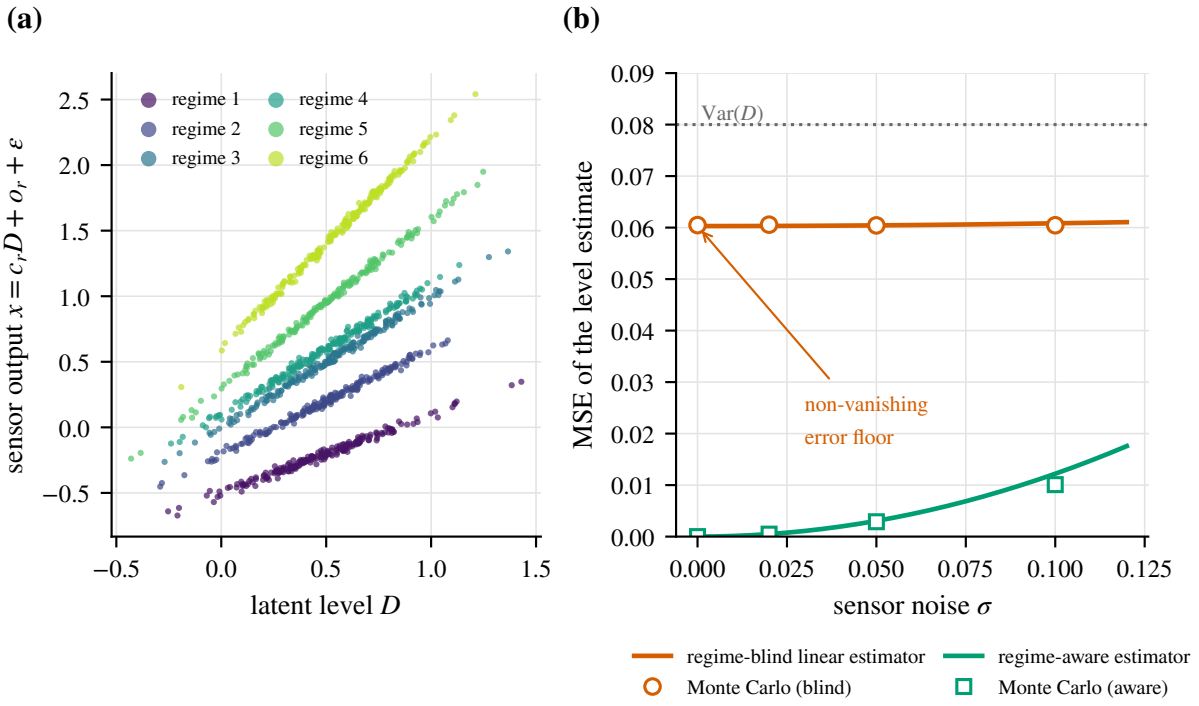


Figure S4: Regime-blind estimation in the six-regime example. (a) Sensor output against latent level for each regime. (b) Error of the regime-blind and regime-aware estimators: formula (lines) and simulation (markers).

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